

Industrial Internship Report on

Python Quiz Game

Prepared by

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Executive Summary

This report presents the work completed during the **Upskill Campus Summer Internship 2026**, organized by **Upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided practical exposure to Python programming and software development through a real-world project.

As part of the internship, I developed a **Python Quiz Game**, a console-based application that allows users to answer multiple-choice questions stored in a JSON file. The application evaluates user responses, calculates scores, displays performance statistics, maintains a high score, and stores player records in a leaderboard.

The project was developed using Python concepts such as file handling, JSON data management, functions, loops, exception handling, modular programming, and user input validation. Additional features such as colored terminal output, random question selection, progress tracking, and performance analysis were implemented to enhance the user experience.

This internship helped me improve my programming skills, logical thinking, debugging techniques, and understanding of software development practices. It also provided valuable experience in using GitHub for version control and project documentation. Overall, the internship strengthened my technical knowledge and increased my confidence in developing real-world Python applications.

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1 Preface

The Upskill Campus Summer Internship 2026 provided an excellent opportunity to gain practical experience in Python programming through a project-based learning approach. During this six-week internship, I enhanced my understanding of programming concepts, problem-solving techniques, and software development practices by developing a real-world application.

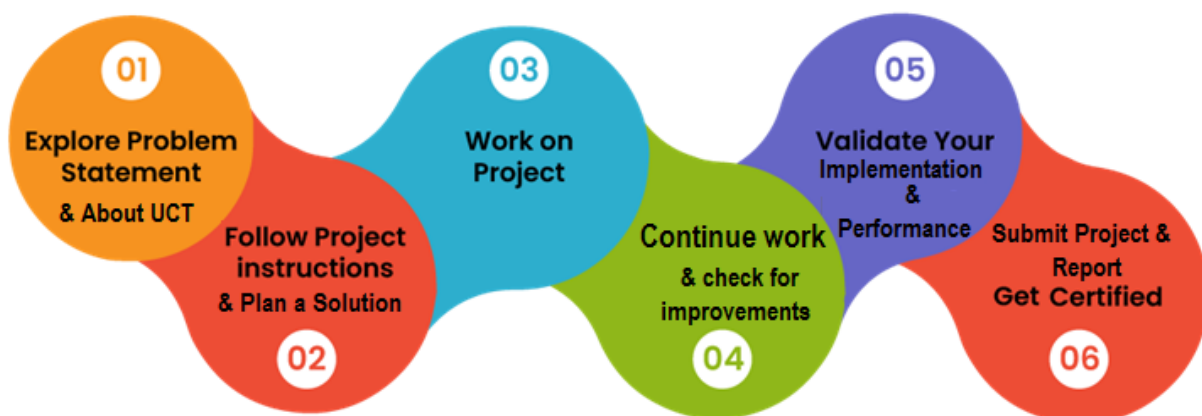
The assigned project, **Python Quiz Game**, focuses on creating an interactive console-based quiz application that allows users to answer multiple-choice questions, calculates scores, and displays performance statistics. The project also includes features such as random question generation, difficulty levels, leaderboard management, high-score tracking, and input validation, making it more user-friendly and reliable.

The internship was well-structured, beginning with learning Python fundamentals, followed by project planning, implementation, testing, documentation, and final deployment using GitHub. This systematic approach helped me understand the complete software development lifecycle.

Throughout this internship, I improved my programming knowledge, debugging skills, file handling techniques, JSON data management, GitHub usage, and documentation skills. These learnings will help me in future academic projects and professional software development.

I sincerely thank **Upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd.**, my college faculty, and my mentors for providing continuous guidance and support throughout this internship.

I encourage my juniors and fellow students to actively participate in internship programs, as they provide valuable practical experience beyond classroom learning and help in developing industry-relevant skills.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



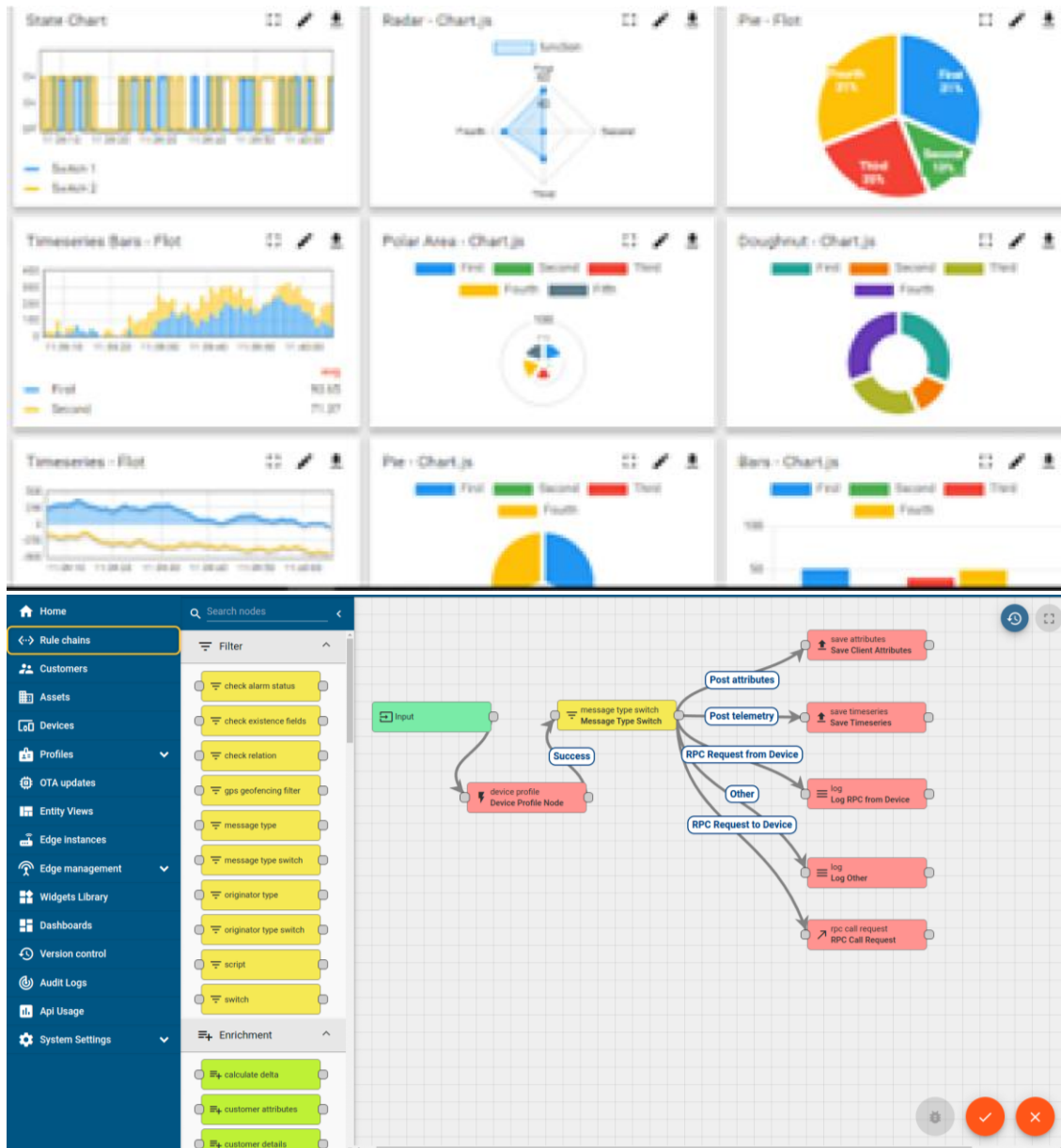
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



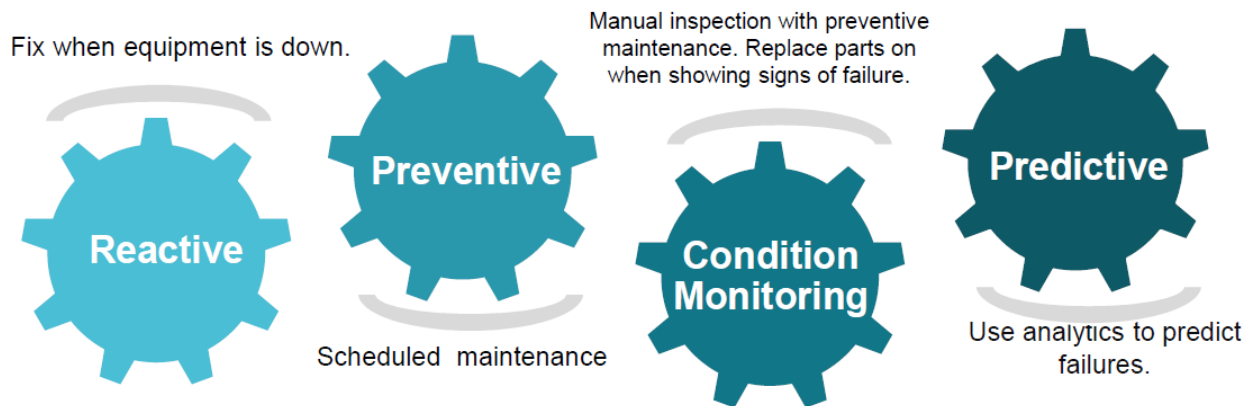


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

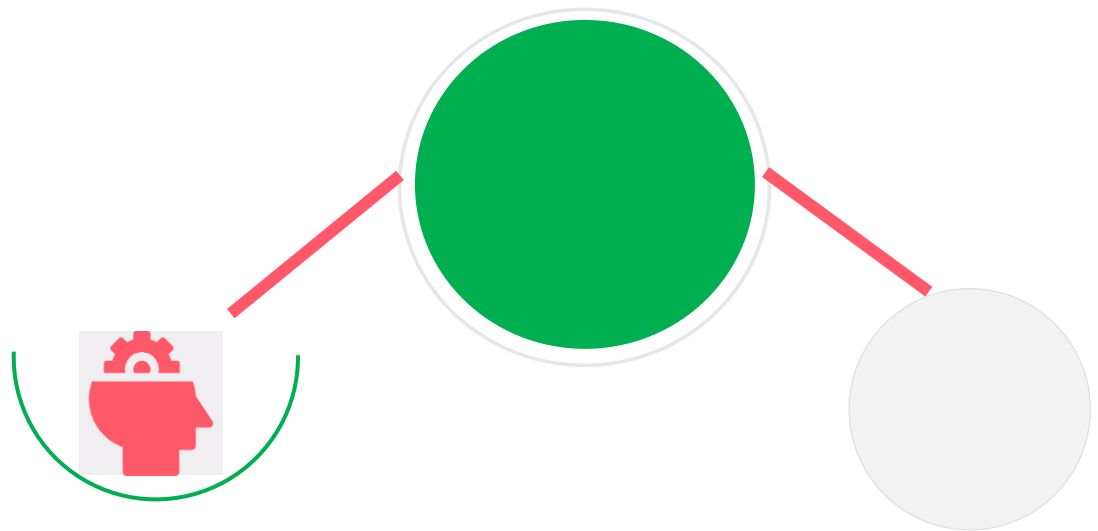
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

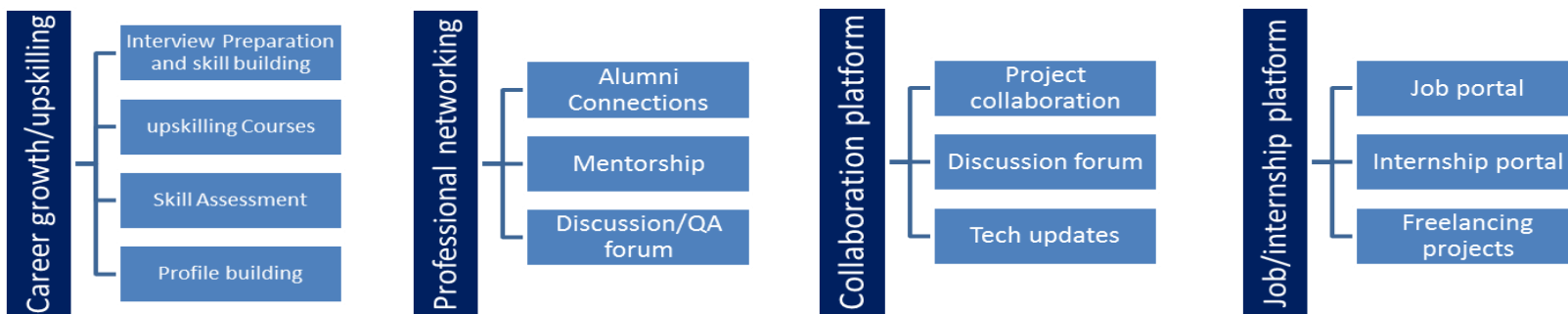
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The main objectives of this internship project are:

- To develop a console-based Quiz Game using Python.
- To strengthen programming skills through practical implementation.
- To understand file handling and JSON data management.
- To implement user input validation and exception handling.
- To calculate quiz scores and analyze user performance.
- To maintain a high score and leaderboard using file handling.
- To gain experience in software development and GitHub version control.
- To improve logical thinking and problem-solving abilities.

2.5 Reference

- [1] Python Official Documentation – <https://docs.python.org/3/>
- [2] Colorama Documentation – <https://pypi.org/project/colorama/>
- [3] GitHub Documentation – <https://docs.github.com/>
- [4] JSON Official Website – <https://www.json.org/>

2.6 Glossary

Terms	Acronym
JavaScript Object Notation	JSON
Integrated Development Environment	IDE
High Score	HS
Leaderboard	LB
Portable Document Format	PDF
Version Control System	VCS
Python Programming Language	Python
Graphical User Interface	GUI

3 Problem Statement

The objective of this project is to develop a **Python-based Quiz Game** that provides an interactive platform for users to test their knowledge through multiple-choice questions.

Many basic quiz applications are either limited in functionality or require an internet connection. The goal of this project is to create a lightweight, user-friendly, console-based application that stores questions in a JSON file and allows users to answer them interactively.

The application validates user input, calculates scores, displays the final percentage, maintains a high score, stores player records in a leaderboard, and provides performance feedback. Additional features such as random question order, difficulty levels, colored terminal output, and progress tracking enhance the overall user experience.

The project demonstrates the practical application of Python programming concepts, including file handling, exception handling, loops, functions, JSON processing, and modular programming.

4 Existing and Proposed solution

- **Existing Solution**

Several quiz applications are available online and offline. However, many of these applications have the following limitations:

- Most quiz applications require an internet connection.
 - Questions cannot be easily modified by users.
 - Limited customization options.
 - No persistent high-score tracking.
 - Basic user interface with minimal feedback.
 - Lack of difficulty levels and progress tracking.
 - Some applications do not provide performance analysis after the quiz.
-

- **Proposed Solution**

The proposed solution is a **Python Quiz Game**, a console-based application developed using Python. It stores quiz questions in a JSON file, allowing easy modification and expansion without changing the program code.

The application provides a user-friendly interface that displays multiple-choice questions, validates user input, calculates scores, and displays a detailed performance summary at the end of the quiz. Additional features such as random question selection, difficulty levels, progress tracking, colored terminal output, high-score management, and a leaderboard improve the overall user experience.

- **Value Addition**

Compared to traditional quiz applications, this project provides several enhanced features:

- User-friendly menu-driven interface.
 - Questions stored in a JSON file for easy maintenance.
 - Randomized question order for every quiz attempt.
 - Difficulty levels (Easy, Medium, Hard).
 - Input validation and exception handling.
 - Colorful terminal interface using the Colorama library.
 - Progress bar to track quiz completion.
 - Automatic score and percentage calculation.
 - High-score management using file handling.
-

- Leaderboard to store and display top player scores.
- Detailed performance analysis after quiz completion.

4.1 Code submission (Github link) :

<https://github.com/viralpatel16/Upskillcampus/blob/main/main.py>

4.2 Report submission (Github link) :

https://github.com/viralpatel16/Upskillcampus/blob/main/QuizGame_Viral_USC_UCT.pdf

5 Proposed Design/ Model

The **Python Quiz Game** follows a modular design in which each component performs a specific task. The application begins by loading quiz questions from a JSON file, then displays a menu for the user. During the quiz, user answers are validated, scores are calculated, and performance statistics are generated. Finally, the application updates the high score and leaderboard before returning to the main menu.

The modular approach makes the application easy to maintain, enhance, and debug.

5.1 High Level Diagram (if applicable)

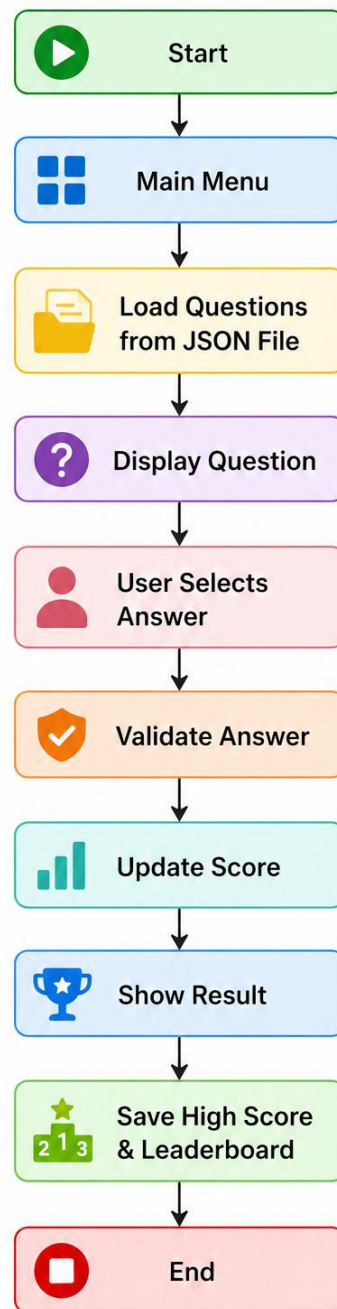


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

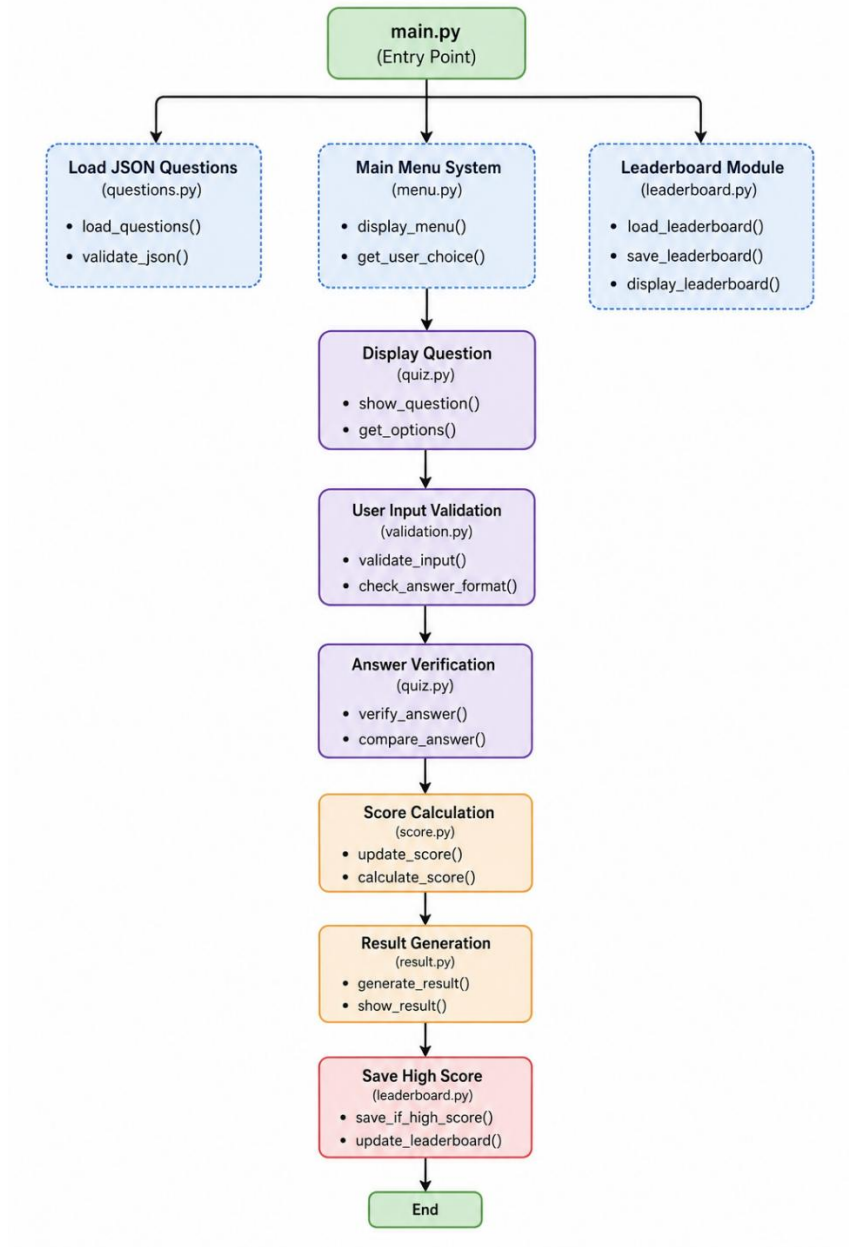


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

User Interface

The project uses a **console-based interface** developed in Python.

The interface provides:

- Main menu for navigation.
- Display of multiple-choice questions.
- Difficulty level display.
- Progress bar indicating quiz completion.
- Color-coded messages for correct and incorrect answers.
- Final score, percentage, and performance summary.
- High score and leaderboard display.

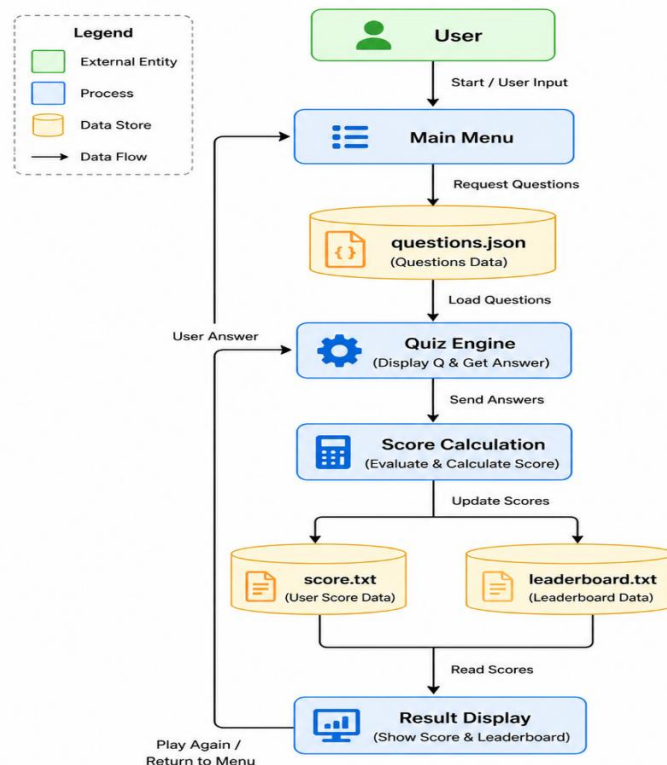


FIGURE 3: DATA FLOW OF THE QUIZ APPLICATION

6 Performance Test

The developed **Python Quiz Game** was tested under different user input conditions to verify its correctness, reliability, and usability. Various functional test cases were executed to ensure the application handled valid and invalid inputs correctly. The performance of features such as question loading, score calculation, leaderboard management, and high-score updates was also verified.

The application uses lightweight JSON and text files for data storage, resulting in fast execution and minimal memory usage. Proper exception handling ensures the program does not terminate unexpectedly due to invalid user inputs.

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	Start Quiz	Quiz starts successfully	Pass
TC-02	Load Questions	Questions loaded from JSON	Pass
TC-03	Enter Correct Answer	Score increases by 1	Pass
TC-04	Enter Wrong Answer	Correct answer displayed	Pass
TC-05	Enter Invalid Input (abc)	Error message displayed	Pass
TC-06	Enter Invalid Option (8)	Input requested again	Pass
TC-07	View High Score	High score displayed	Pass
TC-08	Save New High Score	High score updated	Pass
TC-09	Save Leaderboard	Player score stored	Pass
TC-10	Exit Application	Program terminates normally	Pass

6.2 Test Procedure

The following procedure was followed during testing:

1. Launch the Python Quiz Game application.
2. Select **Start Quiz** from the main menu.
3. Verify that quiz questions are loaded correctly from the JSON file.
4. Answer questions using both correct and incorrect options.
5. Test invalid inputs such as letters and numbers outside the valid range.
6. Verify score calculation and percentage generation.
7. Check that the high score updates correctly when a better score is achieved.
8. Verify that player details are stored in the leaderboard file.
9. View the leaderboard and confirm that scores are displayed in descending order.
10. Exit the application and ensure it closes without errors.

6.3 Performance Outcome

The application performed successfully during all test cases. Question loading from the JSON file was completed instantly, and score calculation was accurate for every quiz attempt. Input validation prevented program crashes caused by invalid user entries. The high-score system and leaderboard worked correctly by storing and retrieving data from text files.

The console-based interface responded quickly, and the use of lightweight file handling resulted in efficient execution with minimal memory usage. Overall, the project achieved all the planned objectives and demonstrated reliable performance under different testing conditions.

6.4 Performance Summary Table :

Parameter	Result
Application Startup	Successful
Question Loading	Successful
Input Validation	Successful
Score Calculation	Accurate
High Score Update	Successful
Leaderboard Storage	Successful
Memory Usage	Low
Execution Speed	Fast
Overall Performance	Excellent

7 My learnings

During the **Upskill Campus Summer Internship 2026**, I gained valuable practical experience in Python programming and software development. Working on the **Python Quiz Game** project helped me strengthen both my technical and problem-solving skills.

The major learnings from this internship are:

- Improved understanding of Python programming concepts such as variables, loops, functions, and conditional statements.
- Learned to use **JSON** for storing and retrieving quiz questions efficiently.
- Gained hands-on experience in **file handling** for maintaining high scores and leaderboards.
- Enhanced knowledge of **exception handling** and user input validation to make applications more reliable.
- Understood the importance of **modular programming** for developing maintainable and reusable code.
- Learned to use external libraries such as **Colorama** to improve the user interface.
- Developed logical thinking and debugging skills while testing different scenarios.
- Gained practical experience in using **GitHub** for version control, code management, and project sharing.
- Improved software documentation skills by preparing a structured internship report.
- Built confidence in developing complete Python applications from planning to implementation and testing.

Overall, this internship provided practical exposure to software development and helped bridge the gap between theoretical knowledge and real-world application development.

8 Future work scope

Although the current version of the **Python Quiz Game** successfully fulfills its objectives, several enhancements can be implemented in future versions to improve functionality and user experience.

Some possible future improvements are:

- Develop a **Graphical User Interface (GUI)** using Tkinter or PyQt.
- Integrate a **database (MySQL or SQLite)** instead of text files for storing user records and quiz history.
- Add **user login and authentication** for personalized quiz sessions.
- Include **multiple quiz categories** such as Science, Technology, Sports, and General Knowledge.
- Provide **different difficulty modes** with adaptive question selection.
- Add a **timer** for each question to make the quiz more challenging.
- Display **graphs and performance analytics** after quiz completion.
- Enable **online multiplayer quizzes** using networking concepts.
- Develop a **web-based version** using Flask or Django.
- Create a **mobile application** using Kivy or another cross-platform framework.

These enhancements would make the application more interactive, scalable, and suitable for educational as well as competitive learning environments.